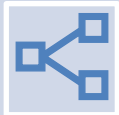


Week 6 Lab: Uncertainty-Aware AI Systems



Train baseline model



Measure confidence



Observe OOD



Design HITL

Dataset



Kaggle Credit
Card Fraud



Binary
classification



Imbalanced
dataset



Fraud vs
Legitimate

Overall Workflow

Load Data

Train Model

Confidence

OOD

Threshold

Trade-offs

Baseline Model

- Logistic Regression
- Measure Clean Accuracy ONLY!

Confidence Scores

- Confidence = Highest Probability
- Example: Fraud=94%
- Should every prediction be trusted?
- Hint: Refer back to our Week 6 Lecture

OOD Samples

- Increase transaction amount dramatically
- Change transaction timing significantly

Silent Failure

- High confidence
- Wrong prediction
- Discuss why

Human-in-the-Loop Example

- $\geq 80\%$ Auto
- $< 80\%$ Human Review
- 80% is example only
- Note: Come up with your own Threshold

Threshold Selection

- Compare 60/75/90%
- Automation vs Safety

Trade-offs

- False Positives
- False Negatives
- Human workload
- Safety

Deliverables

- Threshold
- Trade-offs
- Workflow
- Reflection
- Deadline: July 7th as per your Teams ask